



The
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Question # 1:

—(a)—

Data Preprocessing:

In real-world scenarios, datasets are often incomplete, inconsistent, and noisy. Therefore, preprocessing is a crucial step before applying any data mining technique. The common issues include missing values, duplicate records, and outliers.

1- Handling Missing Values

Missing values occur when some attributes in the dataset are not recorded. These may arise due to human error, system failure or incomplete data collection.

Techniques to handle missing values include:

- Removing the tuple if the number of missing values is small.
- Replacing numerical missing values with mean or median.
- Remove rows or columns with missing values.

2. Handling Duplicate Entries

Duplicate entries refer to repeated records in a dataset. These can distort statistical results and bias machine learning models.

Methods to remove duplicates:

- identifying and removing identical rows.
- Using primary or unique identifiers.
- Using data cleaning tools like python's `panda` Library.

3. Handling Outliers

These are extreme values that differ significantly from others.

- Z-score Methods
- Interquartile Range (IQR)
- Removing outliers if they are caused by errors.

Knowledge discovery in Databases Process

The KDD process is a structured approach to extract useful knowledge from raw data. It consists of following steps:

- Data Selection
- Data Cleaning

- Data Integration
- Data Transformation
- Data Mining
- Pattern Evaluation
- Knowledge Presentation

—(b)—

Role of Variance and Standard Deviation

Variance and Standard deviation measures how spread out data values are:

- Variance: average of squared differences from the mean.
- Standard Deviation: Sq. root of variance (more interpretable)

Role in data analysis:

- Helps understand data dispersion
- Detects outliers
- Used in probability and Statistics...

Types of Classification:

- Binary:
Two possible classes

e.g

Email $\rightarrow$ Spam or not spam

- Multiclass

More than two classes

e.g

- Handwritten digit (0-9)

- Multi-label

An instances can belong to multiple classes at the same time.

- A movie can be both action & comedy.

(c)

Association Rule Mining:

Association rule mining finds relationships between variables in large datasets.

e.g If a customer buys bread, they also buy butter.

Measures:

Support $\rightarrow$ shows how frequently an itemset appears in dataset

$$\text{Support}(A \rightarrow B) = \frac{\text{Transactions containing A and B}}{\text{Total Transactions}}$$

Confidence

Confidence shows how often B appears when A appears.

$$\text{Confidence}(A \rightarrow B) = \text{Support}(A \text{ and } B) / \text{Support}(A)$$

Lift

Lift measures the strenght of the rule.

$$\text{Lift}(A \rightarrow B) = \text{Confidence}(A \rightarrow B) / \text{Support}(B)$$

If $\text{lift} > 1$, items are positively related.

Question # 2

Generating Candidate 3-itemsets

Given frequent 2-itemsets : $L_2 = \{AB, AC, BC\}$

Using Apriori principal:

Step 1: Join step

Combine itemsets with common items:

- $AB \text{ and } AC \rightarrow ABC$
- $AB \text{ and } BC \rightarrow ABC$
- $AC \text{ and } BC \rightarrow ABC$

So candidate 3-itemset, $C_3 = \{ABC\}$

Step 2: Pruning step

Check if all subsets of ABC are frequent: Subsets are:

- AB
- AC
- BC

Since all are in L_2 , ABC is not pruned.

Final candidates 3-itemset, $L_3 = \{ABC\}$

Apriori Algorithm

Apriori is used to find frequent itemsets

Steps

- 1- Find Frequent 1-itemsets
- 2- Generate candidate 2-itemsets
- 3- Prune itemsets below minimum support.
- 4- Repeat until no new itemsets are found.

Limitation

- Requires multiple database scans
- High computation cost

Comparison with FP-Growth

FP-Growth

- Uses FP-Tree structure
- No candidate generation
- Faster than Apriori
- Requires only two database scans.

Question # 3

Supervised Classification in data Mining

Supervised classification is a technique where models are trained using labeled data. The goal is to predict class labels for new data.

Role of python and OOP

Python is widely used in machine learning because of libraries like:

- Numpy
- Pandas
- Scikit-Learn

OOP helps in:

- Creating reusable models
- Organizing code into classes & objects.

Algorithms

1- Decision Tree

- Based on splitting data using conditions
- Uses entropy & information gain

2- Naive Bayes

- Based on probability theory.
- Assumes features are independent.

3- K-Nearest Neighbors (KNN)

- Based on distance metrics.
- Classifies data based on nearest neighbors.

4- Support Vector Machine (SVM)

- Uses hyperplanes to separate classes.
- Maximizes margin b/w classes.

Strength and Limitation

Decision Tree :

Strength → Easy to interpret

Limitation → Can overfit

Naive Bayes:

Strength → Fast & simple

Limitation → Assumes independence.